NATURAL LANGUAGE PROCESSING | RESEARCH AND IMPLEMENTATION PROJECT

RegulaTrace Compliance Obligation Intelligence System

Research Proposal and Implementation Blueprint for Evidence Linked Regulatory Analysis

A hybrid compliance compiler that converts changing legal text into atomic obligations, validated evidence graphs and reviewer-ready policy-gap signals.

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PROJECT STATUS An executable rule-based sentence classifier, heuristic field extractor and 30-example synthetic evaluation are the demonstrated baseline. The evidence graph, change radar, policy comparator, learned models and review workspace are research-backed target modules and are not represented as deployed.

Executive Summary

RegulaTrace is designed to solve the first hard problem in compliance operations: turning long, changing and exception-heavy regulatory text into structured requirements that remain linked to their exact source. The system is not a chatbot that produces legal answers from memory. It is a controlled pipeline that compiles legal provisions into reviewable obligation records, validates their structure, maps them to policies and controls, and records every transformation and human decision.

The central contribution is an evidence-linked obligation graph. Each obligation is represented as an actor, modality, action, object, trigger, deadline, condition, exception, scope and authority, with immutable citations and field-level confidence. This creates the foundation for four high-value capabilities: obligation inventory generation, policy-gap triage, regulatory change impact analysis and audit-ready evidence packaging.

The design combines transparent rules, legal-domain classifiers, span extraction, temporal normalization, hybrid retrieval, natural language inference, graph constraints and human review. Recent work shows why this hybrid approach is necessary: regulatory information extraction remains data-scarce; multi-passage retrieval is materially harder than single-passage retrieval; and even domain-specific fine-tuning can fail to improve regulatory variation detection [6-10]. The proposed architecture therefore requires strong baselines, document-level evaluation, calibrated abstention and explicit evidence rather than model confidence alone.

What makes the project defensible and distinctive

- Atomic obligation compiler: converts a clause into a machine-checkable intermediate representation without discarding the original words.
- Regulatory change radar: detects added, removed and semantically altered duties, then traces which policies, controls, owners and evidence may be affected.
- Constraint-aware policy comparison: checks polarity, actor, timing, thresholds, scope and exceptions after retrieval instead of treating semantic similarity as compliance proof.
- Proof-carrying outputs: every extracted field and gap signal carries a source span, document version, transformation history, confidence and reviewer decision.
- Measured human benefit: evaluation includes critical-miss rate, citation integrity and reviewer time saved, not only model accuracy.

Recommended academic deliverable

Build a convincing vertical slice rather than a shallow enterprise imitation: official-source ingestion for two regulations, clause extraction, the obligation schema, one policy-comparison workflow, one change-impact demo, a reviewer screen and a document-split evaluation. This is ambitious enough to be memorable and constrained enough to be implemented and defended in a semester.

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Reading note External research was checked through 23 October 2026. Numeric results from the original 30-example baseline are retained as a sanity check, not presented as real-world legal performance.

1 Strategic Vision

From isolated clause extraction to a living map of what an organization must do and why.

The problem is not merely finding the word must. A useful compliance system must know which entity is bound, what action is required or prohibited, what event starts the clock, how long the deadline lasts, what conditions limit the duty, which exceptions apply, which version is authoritative and what internal process claims to satisfy it. If any of those pieces is detached from its source, the output is difficult to defend.

RegulaTrace treats regulatory analysis as a compilation problem. Human-authored legal text is the source language. The obligation record is an intermediate representation. Policy mappings and control tests are downstream analyses. Reviewer confirmation is the release gate. This framing supports reproducibility, testing and change impact in a way that a one-shot summary cannot.

North star outcome

When a regulation changes, the system should identify the affected obligation, show the old and new evidence, locate candidate internal policies, compare material constraints, identify responsible owners and assemble a review package. It should also know when the evidence is insufficient and stop.

Primary users

USER	DECISION SUPPORTED	OUTPUT
Compliance analyst	What duties apply and what requires review	Evidence-linked obligation register and exception queue
Policy owner	Which internal document may need revision	Constraint differences with cited source and suggested review questions
Control owner	Which operational control and evidence support the duty	Owner-control-evidence links and stale-evidence alerts
Auditor	How a conclusion was reached	Versioned source, extraction history, reviewer rationale and exportable audit package
Research team	Where the model fails	Document-split metrics, calibration curves and tagged error catalogue

Design principles

1. Evidence before inference. Preserve official source identifiers, hierarchy, page or section anchors and exact spans before predicting anything.
2. Decompose before comparing. Compare actor, action, polarity, timing, conditions and exceptions independently before forming a review signal.
3. Abstention is a feature. Unknown, inherited or cross-referenced information remains unresolved until evidence is found or a reviewer confirms it.
4. Version everything. Legal source, model, rules, schema, policy and reviewer decisions must be time-stamped and reproducible.
5. Optimize for critical misses. A missed high-severity duty is not equivalent to an extra item in a review queue.

2 Project Contribution and Novelty

A serious project is differentiated by the structure of its contribution, not by inflated claims.

The proposed system moves beyond a conventional obligation extractor in five connected ways.

CONTRIBUTION	WHAT IT ADDS	WHY IT MATTERS
Obligation intermediate representation	A typed record for modality, actors, action, object, timing, triggers, exceptions, scope, authority and provenance	Enables validation, comparison, graph storage and repeatable evaluation
Evidence and impact graph	Links provisions to obligations, policies, controls, owners, evidence, reviews and revisions	Turns extraction into operational change management rather than a static spreadsheet
Hybrid policy comparator	Retrieval followed by NLI and explicit constraint checks	Reduces the risk of calling topical similarity a compliance match
Change radar	Structure-aware alignment and semantic diff between regulatory versions	Shows which downstream records may be stale when law or policy changes
Assurance layer	Schema validation, provenance, field confidence, abstention and review state	Makes each output inspectable and suitable for a high-stakes workflow

Positioning against common alternatives

CAPABILITY	KEYWORD EXTRACTOR	GENERIC RAG CHATBOT	REGULATRACE TARGET
Exact source provenance	Partial	Depends on prompting	Mandatory and versioned
Conditions and exceptions	Brittle patterns	May summarize incorrectly	Structured fields plus scope checks
Policy alignment	No	Similarity and generated explanation	Retrieval plus NLI plus constraint comparison
Regulatory change impact	No	Manual re-query	Revision graph and affected-record queue
Uncertainty handling	Rule hit or miss	Often overconfident	Field confidence abstention and reviewer state
Evaluation	Sentence accuracy	Answer quality	End-to-end evidence and critical-miss metrics

Research claim

The research question is whether an evidence-first hybrid pipeline can produce more complete, reviewable and change-aware obligation records than either lexical rules or an unconstrained language model, while reducing analyst effort without increasing critical misses.

3 Research Foundation

The architecture is grounded in legal NLP benchmarks, regulatory information extraction, document standards and AI assurance.

Legal-domain pretraining and benchmarks

LEGAL-BERT and LexGLUE show that domain-adapted representations can improve legal-language tasks, while CUAD and ContractNLI demonstrate the value of expert labels, evidence spans and document-level reasoning [1-4]. LegalBench broadens evaluation beyond one aggregate notion of legal reasoning [9]. These resources are useful comparators, but none is a complete benchmark for this project's obligation schema.

Directly relevant regulatory NLP

Low-resource work on EU legislation reports the difficulty of deontic classification across obligations, permissions, prohibitions, rights and powers [5]. In 2025, Piskorski and Skotarczak introduced a structured information-extraction task for information-provision requirements in EU law, including providing entity, target entity, deadlines and frequency [6]. Their experiments are especially important because a character n-gram SVM reached 0.927 F1 on requirement detection and outperformed the reported legal transformer on that small dataset, reinforcing the need for classical baselines rather than assuming a larger model will win.

Regulatory retrieval and dispersed evidence

RIRAG 2025 used ObliQA for regulatory passage retrieval and answer generation [7]. ObliQA-MP later focused on questions whose support is distributed across multiple passages; its authors report a notable performance drop from single- to multi-passage retrieval and gains from combining lexical, semantic and graph-derived signals [8]. RegulaTrace therefore preserves document hierarchy and cross-references and evaluates evidence retrieval separately from answer generation.

Standards for durable legal data

Akoma Ntoso Version 2.0 Part 2 AKN 3.1 became an OASIS Standard effective 2 October 2026, providing a current structural reference for legal documents [11]. LegalRuleML provides concepts for representing legal rules, deontic operators, violations and penalties [12]. W3C PROV-O supports interoperable provenance; SHACL supports graph validation; and ISO 8601 supports unambiguous dates, durations and recurring intervals [13-15]. The project does not need to implement each standard in full, but it should align its identifiers and data model with them.

AI assurance and governance

NIST AI RMF and its Generative AI Profile emphasize lifecycle risk management, evaluation, transparency and governance [16-17]. The EU AI Act illustrates the kind of actor-specific, documentation-heavy obligation set the system should be able to structure, while GDPR Article 33 remains an ideal test for anchored deadlines and conditional exceptions [19-20].

Research synthesis

EVIDENCE	DESIGN CONSEQUENCE
Legal tasks vary and legal-domain models do not solve every task	Evaluate multiple baselines and task-specific failure modes
Regulatory support may span several sections	Retain hierarchy cross-references and multi-passage retrieval
Small datasets can favor simple models	Keep rules TF-IDF and character n-gram baselines
Similarity can miss logical incompatibility	Compare structured constraints after retrieval
Standards already exist for legal structure provenance and validation	Use stable identifiers provenance events and schema gates
High-stakes outputs require governance	Require reviewer confirmation and record reasons and uncertainty

4 Scope Requirements and Success Criteria

The submission should be ambitious at the system level and precise about what the prototype actually proves.

Project scope

IN SCOPE FOR VERTICAL SLICE	OUT OF SCOPE OR DEFERRED
English regulations and organizational policies	Autonomous legal interpretation or certification
Born-digital PDF DOCX text and official XML feeds	Production OCR across arbitrary scans
Clause hierarchy cross-references and version metadata	Complete multilingual and multi-jurisdiction support
Five-way modality plus actor action timing trigger condition exception	All legal concepts rights powers remedies and penalties
Policy candidate retrieval and review signals	Automatic proof that an organization is compliant in practice
One end-to-end change-impact demonstration	Enterprise integrations and continuous production monitoring

Functional requirements

ID	REQUIREMENT	TESTABLE OUTPUT
FR 01	Ingest source and preserve identity version hierarchy and location	Re-open any extracted record at exact supporting text
FR 02	Detect candidate normative clauses and classify modality	Five-way label plus score and rule or model trace
FR 03	Extract typed spans and normalize time without inventing absent fields	Atomic obligation record with null or unresolved states
FR 04	Validate records before storage	Schema report and blocking errors for missing provenance
FR 05	Retrieve and compare policy evidence	Ranked passages and per-constraint result
FR 06	Detect revisions and downstream impact	Added removed changed obligations and affected graph nodes
FR 07	Support analyst correction and approval	Versioned decision with rationale and owner
FR 08	Export machine and human readable results	JSON CSV and audit report

Non-functional requirements

- Traceability: 100 percent of stored obligations must contain immutable source identity, version and evidence span.
- Privacy: local-first inference must be available for sensitive policies; no document leaves the environment without explicit configuration.
- Reproducibility: rule set, model, prompt, schema and dataset versions are recorded for every run.
- Accessibility: status must not be color-only; evidence and decisions must be readable and exportable.
- Performance: interactive review should return a clause and its evidence within two seconds on the demonstration corpus after indexing.
- Safety: no automated conclusion uses the labels compliant or non-compliant without human confirmation and evidence scope.

Prototype acceptance targets

MEASURE	MINIMUM TARGET	RATIONALE
Candidate clause recall	>= 0.95 on held-out documents	Missing a duty is costlier than reviewing an extra candidate
Modality macro F1	>= 0.80	Balances minority classes
Actor action relaxed span F1	>= 0.75	Supports useful structured review
Deadline plus anchor accuracy	>= 0.85 on applicable cases	A duration without its trigger is unsafe
Exception preservation	>= 0.90 recall	Exceptions can reverse applicability
Provenance integrity	1.00	Every record must be auditable
Policy evidence recall at 10	>= 0.90	Comparator cannot reason over evidence it did not retrieve
Critical false-negative rate	<= 0.02 on challenge set	Primary safety metric
Median reviewer time	>= 30 percent reduction vs manual baseline	Measures actual usefulness

5 System Architecture

The platform separates acquisition, understanding, reasoning, review and assurance so every stage can be tested.

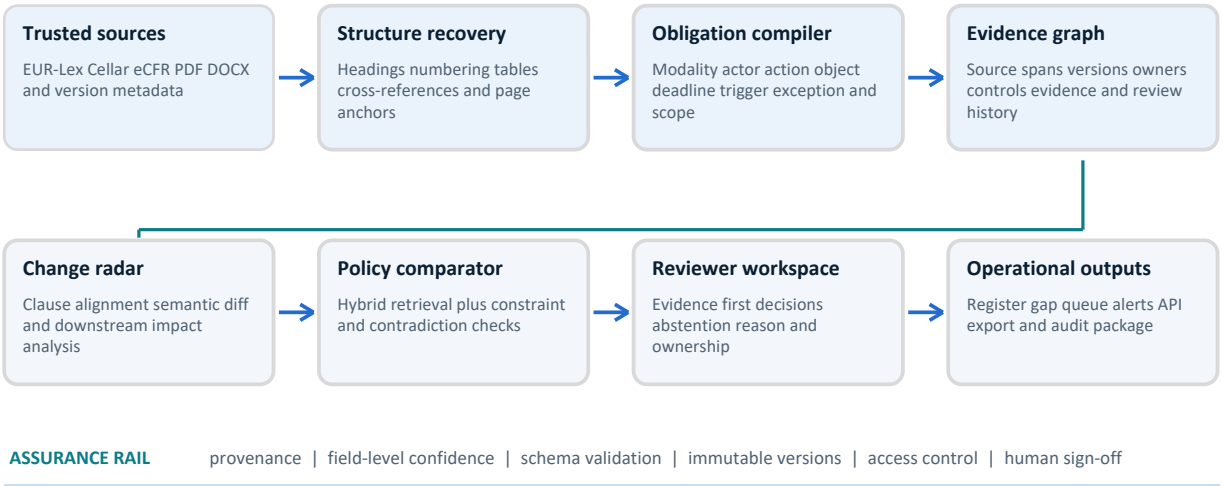


Figure 1 Target architecture and assurance rail

The architecture is intentionally hybrid. Deterministic components handle file identity, hierarchy, source anchors, hashing and schema validation. Statistical models identify language patterns and rank evidence. Constrained generation may propose structured fields or rationales, but outputs are accepted only after span verification and validation. The graph connects every downstream decision to the source and makes revision impact computable.

Stage responsibilities

STAGE	PRIMARY METHOD	FAIL SAFE
Source acquisition	Official feeds uploads metadata hashing	Reject duplicate or unverifiable version
Structure recovery	XML hierarchy PDF layout headings list parser	Keep raw page and block coordinates
Candidate detection	Rules plus high-recall classifier	Send uncertain normative text to review
Field extraction	Span model dependency rules constrained LLM	Null or unresolved instead of guessed value
Normalization	Controlled vocabularies ISO time representation	Store original expression beside normalized value
Graph construction	Typed nodes edges provenance and SHACL checks	Block incomplete provenance
Policy retrieval	BM25 dense retrieval and rank fusion	Show all cited candidates and retrieval scores
Constraint comparison	NLI plus symbolic checks	Insufficient evidence when dimensions cannot be aligned
Change impact	Structure-aware diff graph traversal	Never auto-close affected obligations
Review and export	Analyst decision audit trail	Require reason for override and rejection

6 Obligation Representation and Evidence Graph

The data model is the project's core intellectual asset because it determines what can be reasoned about and audited.

Atomic obligation model

One sentence may contain several duties, and one duty may depend on text in several provisions. RegulaTrace therefore stores an obligation as an atomic rule with explicit support spans rather than treating a sentence as the final unit.

FIELD GROUP	FIELDS	VALIDATION RULE
Identity	obligation id document id version jurisdiction authority section	Stable identifier and resolvable source required
Normative core	modality actor action object recipient	Modality required actor may be explicit inherited or unresolved
Applicability	scope subject class product geography threshold	Original qualifying span retained
Temporal	trigger anchor deadline duration cadence effective dates	Normalized value never replaces raw text
Logic	conditions exceptions cross-references alternatives	Scope relation must be explicit or unresolved
Operational	owner policy control evidence risk severity	Reviewer-created links separated from source-extracted facts
Assurance	source spans extractor version field confidence review state rationale	No confirmed state without reviewer identity and time

Evidence graph

A compliance record is a connected chain of legal source, extracted meaning, operational response, evidence and review.

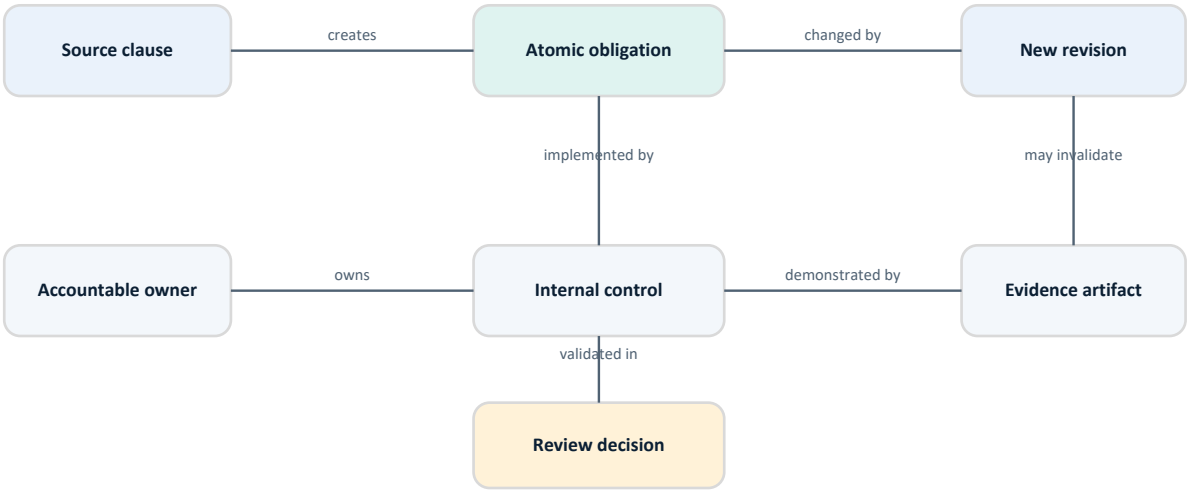


Figure 2 Minimum evidence and impact graph

The graph supports questions a flat spreadsheet cannot answer reliably: Which policies depend on a changed clause? Which obligations have no owner? Which controls have stale evidence? Which reviewer overrode a model prediction? Which duties inherit an actor from a parent provision? PROV-O concepts can describe derivation and responsibility, while SHACL-style constraints can reject structurally incomplete records [13-14].

Example JSON intermediate representation

```
{
  "obligation_id": "EU-GDPR-33-1-A",
  "actor": {"text": "controller", "status": "EXPLICIT"},
  "modality": "OBLIGATION",
  "action": "notify supervisory authority",
  "trigger": "becoming aware of a personal data breach",
  "deadline": {"raw": "where feasible not later than 72 hours", "max_hours": 72},
  "exception": "unless the breach is unlikely to result in risk",
  "source": {"celex": "32016R0679", "article": "33(1)"},
  "review_state": "UNREVIEWED"
}
```

The example is a structured paraphrase for system design, not a substitute for the official text or legal interpretation [20].

7 NLP and Reasoning Methodology

Each model is assigned a narrow responsibility, compared with a strong baseline and prevented from silently changing the source.

7.1 Structure-aware preprocessing

The parser preserves document hierarchy before sentence splitting. Numbered paragraphs, list items, definitions, tables, headings and cross-references remain addressable. Born-digital XML from official sources should be preferred where available; PDF layout extraction is a fallback. Akoma Ntoso identifiers are a useful model for stable legal references, while official EUR-Lex Cellar and GovInfo bulk data provide authoritative machine-readable acquisition paths [11, 21-22].

7.2 Candidate detection and modality

A high-recall candidate detector combines ordered lexical rules, character n-grams and a fine-tuned legal encoder. The five operational classes remain obligation, prohibition, recommendation, permission and information. A sixth routing state, ambiguous, is used when the system cannot distinguish deontic from epistemic language or when scope depends on another provision.

BASELINE	PURPOSE	EXPECTED STRENGTH
Ordered lexical rules	Transparent lower bound and edge-case tests	Explicit shall must should may patterns
Character n-gram linear model	Strong small-data comparison	Robustness to drafting variants and morphology
Legal-domain encoder	Contextual sentence or clause classification	Implicit necessity and contextual modality
Long-document encoder or hierarchy model	Use parent heading and surrounding provisions	Inherited actors cross-references and long scope
Constrained language model	Optional structured proposal and explanation	Complex paraphrase but must pass evidence validation

7.3 Span extraction

A multi-task span model proposes actor, action, object, recipient, trigger, deadline, frequency, condition and exception. Dependency rules supply interpretable candidates and handle direct patterns. A constrained model can fill difficult slots, but every non-null value must be supported by an exact source span or explicitly marked inferred with its support chain.

7.4 Temporal and exception reasoning

Time expressions are stored as raw text plus normalized structure: amount, unit, comparator, recurrence and anchoring event. 'Within 72 hours' is incomplete unless the event that starts the period is captured. Domain-sensitive temporal tagging and ISO 8601 provide foundations, but legal deadlines also need business-day rules, effective dates, extensions and relative anchors [15, 24]. Exception detection must resolve which duty or condition an unless, except or provided that phrase modifies.

7.5 Confidence and selective prediction

Confidence is field-specific, not a single probability for the whole record. The system should calibrate predictions on held-out documents and use selective prediction: confirm only above a validated threshold, queue the middle range and abstain when evidence is missing or distribution shift is detected. Conformal prediction can be evaluated as a way to return label sets with target coverage rather than a single overconfident class [25].

7.6 Validation

- Reject records without source identity, version and evidence span.
- Warn when a deadline is present without a trigger or when a recurring frequency lacks a unit.
- Require a polarity check when negation and modal language occur in different spans.
- Preserve cross-reference targets and mark unresolved references as blocking issues.
- Keep source-extracted facts separate from analyst-created ownership and control mappings.

8 Policy Comparison and Regulatory Change Intelligence

The differentiator is not retrieval alone but disciplined comparison and downstream impact.

Policy comparison

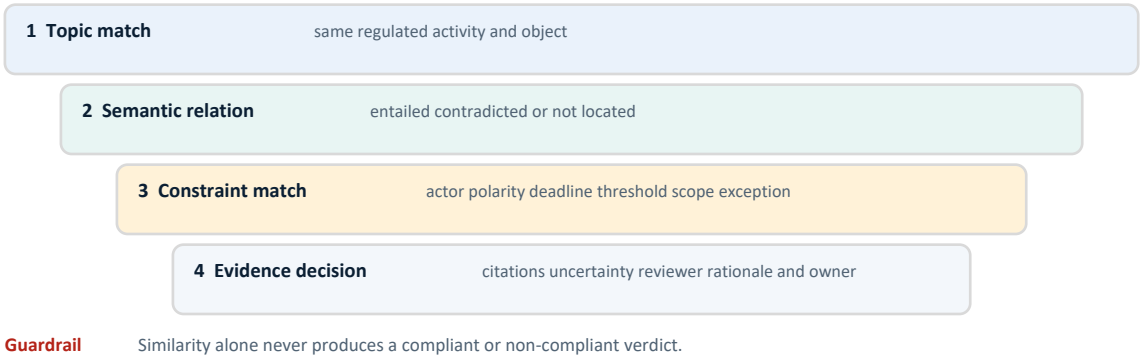


Figure 3 Layered policy-comparison decision

A comparison begins with high-recall retrieval, combining lexical BM25 and dense semantic candidates. A reranker considers headings, actors, control topics and graph proximity. NLI then estimates entailment, contradiction or not-mentioned, but a deterministic constraint layer separately checks modality, deadlines, thresholds, exceptions and scope. The final state is one of likely aligned, potential mismatch, insufficient evidence or not applicable, each with cited text and reviewer rationale.

Comparison dimensions

DIMENSION	QUESTION	EXAMPLE SIGNAL
Actor	Does the policy bind the relevant entity or role	Controller duty mapped only to vendor procedure
Action and object	Is the same act performed on the same subject	Review accounts vs review applications
Polarity	Is a prohibition weakened into permission	Must not disclose vs may disclose
Timing	Are cadence deadline and trigger compatible	Annual review vs at least every 90 days
Threshold	Is the numeric or qualitative threshold at least as strict	More than 500 records vs any breach
Condition	Are prerequisite events equivalent	After awareness vs after confirmation
Exception	Does policy add an unsupported exclusion	Except contractors when law covers all staff
Scope	Are jurisdiction product data class and process aligned	EU customers vs all customers

Regulatory change radar

The change radar compares two authoritative versions in four passes: structural alignment, sentence or clause alignment, semantic change classification, and graph impact propagation. It labels additions, removals, moved text, definition changes, tightened or relaxed constraints, changed deadlines, modified exceptions and cross-reference updates. Regulatory variation research shows that alignment remains difficult and that domain fine-tuning is not automatically beneficial, so the prototype must measure alignment quality and surface uncertain pairs rather than hiding them [10].

1. Acquire both source versions and verify identifiers, effective dates and hashes.
2. Align hierarchy using stable references, headings, numbering and lexical anchors.
3. Classify material semantic differences at the obligation-field level.
4. Create a revision event and traverse affected policies, controls, owners and evidence.
5. Prioritize the review queue by severity, confidence, effective date and operational coverage.

Impact severity model

A transparent triage score can combine legal criticality, change magnitude, implementation coverage, time to effective date and extraction uncertainty. The score ranks review work; it does not estimate legal liability. Each component and weight remains visible to the reviewer.

9 Prototype Design and Technical Stack

The recommended prototype is a vertical slice that can be demonstrated locally, inspected in code and evaluated end to end.

Reference implementation

LAYER	RECOMMENDED COMPONENT	WHY
Ingestion	Python connectors for local files EUR-Lex and GovInfo samples	Reproducible and source-aware
Parsing	PyMuPDF python-docx XML parser plus layout metadata	Covers common academic demo formats
NLP	spaCy or Stanza rules plus Hugging Face models	Interpretable hybrid pipeline
Retrieval	BM25 plus sentence embeddings and rank fusion	Balances exact legal language and paraphrase
Storage	PostgreSQL JSONB with pgvector or SQLite for demo	Typed records versioning and portable deployment
Graph	RDFLib or a graph database adapter	Evidence and change-impact traversal
Validation	JSON Schema plus SHACL-style graph constraints	Blocks malformed or untraceable records
API	FastAPI	Typed local endpoints and automatic schema documentation
Reviewer UI	Streamlit for rapid prototype or React for full build	Side-by-side evidence correction and approval
Experiment tracking	Git DVC or MLflow plus fixed seeds	Repeatable datasets models and metrics

Repository structure

```
src ingestion parsing extraction temporal graph comparison change api
schemas obligation.schema.json policy-mapping.schema.json graph-shapes.ttl
data raw interim annotations splits challenge-set
tests unit adversarial regression provenance
app reviewer interface and demonstration fixtures
reports evaluation cards failure catalogue and model card
```

Minimum demonstration endpoints

ENDPOINT	BEHAVIOUR
POST documents	Ingest file metadata hash hierarchy and version
POST extract	Return candidate clauses and structured obligation records
GET obligations id evidence	Return exact supporting spans and transformation trace
POST compare policy	Return ranked evidence and dimension-level comparison
POST revisions compare	Return aligned clauses semantic changes and impacted records
POST reviews	Save confirmation correction rejection rationale and reviewer
GET exports audit	Generate obligation register and evidence package

Local-first operating mode

The default demonstration can run without sending regulatory text or internal policies to an external API. An optional model-provider adapter may be evaluated separately, but results must be labeled by provider, model, date, prompt and privacy configuration. Recent 2026 work on sensitive legal information extraction reinforces the practicality of comparing local and hybrid approaches instead of assuming a remote large model is required [26].

10 Data and Annotation Strategy

The project will be judged by the quality of its task definition and test data as much as by the model.

Corpus design

The recommended corpus has three layers. Layer A preserves the original 30 synthetic sentences as regression tests. Layer B contains 1,000 to 2,000 clauses sampled from a small number of official, versioned instruments, including hard negatives and cross-references. Layer C is a focused policy-alignment and change-impact set with paired regulations, internal-style policy statements and revision pairs. Public legal sources must be stored under their applicable terms; proprietary policy text should be replaced with authorized or synthetic material for the academic submission.

Source families

SOURCE FAMILY	ROLE IN EVALUATION	EXAMPLE
Data protection	Deadlines triggers risk exceptions and notifications	GDPR Article 33 and related provisions
AI governance	Multiple actors documentation logging and lifecycle duties	EU AI Act Articles 9 13 and 16
Cybersecurity framework	Non-legal control and governance mapping	NIST CSF 2 0 Govern function
US federal regulation sample	Different drafting system and official XML	Selected GovInfo or eCFR XML title
Synthetic internal policy	Controlled gap-analysis ground truth	Cadence threshold scope and exception variants

Annotation workflow

1. Write a field-by-field annotation guide with positive examples, counterexamples and unresolved states.
2. Train two annotators on a pilot set and revise the guide before scaling.
3. Double-annotate at least 20 percent of the corpus and adjudicate disagreements.
4. Measure agreement separately for modality, span boundaries, temporal anchor and exception scope.
5. Split by source document or regulator, never by near-duplicate sentence.
6. Freeze a hidden challenge set covering negation, passive voice, inherited actors, nested exceptions, cross-references and tables.

Core annotation rules

RULE	DECISION
Shortest faithful span	Select the smallest actor or action span that preserves legal meaning
No guessing	Use null or unresolved for absent actor deadline or target
Inherited context	Mark inferred and cite the parent heading definition or antecedent
Negation	Attach it to modality or action according to scope and preserve exact words
Temporal anchor	Annotate both duration and starting event
Exception scope	Record which obligation or condition the exception limits
Multiple duties	Create separate atomic records linked to the same provision
Cross-reference	Store target and unresolved state when target content is unavailable

11 Evaluation and Experimental Plan

The evaluation is designed to show where the system is trustworthy, where it abstains and whether it saves meaningful analyst time.

Existing baseline

The initial project reports 21 correct classifications out of 30 synthetic sentences, or 70.0 percent accuracy, with approximate macro F1 of 0.75. This is retained as a reproducible lexical sanity check. It is not an estimate of performance on legal documents, and it does not evaluate field extraction, cross-references or policy matching.

REFERENCE CLASS	N	PRECISION	RECALL	F1
Obligation	12	0.80	0.67	0.73
Prohibition	5	0.75	0.60	0.67
Recommendation	4	1.00	0.75	0.86
Permission	3	1.00	1.00	1.00
Information	6	0.40	0.67	0.50

Model comparison

Run the same document-split data through majority, lexical rules, character n-gram linear classifier, general encoder, legal-domain encoder and hybrid model. If a language model is used, compare zero-shot, few-shot schema-constrained and retrieval-grounded variants. The hybrid is successful only if gains survive significance testing and error analysis.

Evaluation matrix

TASK	PRIMARY METRIC	SAFETY OR DIAGNOSTIC METRIC
Candidate detection	Recall and precision	Critical miss rate by severity
Modality	Macro F1 and per-class recall	Confusion by negation modality sense and source
Field extraction	Exact and relaxed span F1	Unsupported-field rate
Temporal	Normalized value plus anchor accuracy	Deadline without anchor rate
Exceptions	Span and scope accuracy	Exception drop rate
Retrieval	Recall at k MAP and nDCG	Evidence diversity and citation coverage
Policy comparison	Macro F1 by outcome	Serious missed mismatch count
Change detection	Alignment and material-change F1	Downstream impact recall
Calibration	Expected calibration error Brier score	Selective risk vs coverage
Human use	Median time and correction count	Agreement confidence and error severity
Provenance	Integrity pass rate	Broken source-link count

Ablation studies

- Remove parent headings and cross-references to quantify the value of hierarchy.
- Compare dense retrieval, BM25, fusion and graph-aware reranking.
- Remove explicit constraint checks to measure how often NLI misses timing or exception conflicts.
- Replace field-level confidence with one record score to test whether uncertainty becomes less useful.
- Compare raw language-model JSON with span-verified and schema-validated output.

Human-centered study

Use a within-subject design with a small group of trained reviewers. Each reviewer completes matched tasks manually and with RegulaTrace. Measure time, missed fields, unsupported conclusions, number of source openings, perceived workload and trust calibration. Reviewers must be able to reject system output; disagreement is recorded as research data rather than treated as user error.

12 Demonstration Case Study

A memorable demonstration follows one obligation from official source to policy decision and then shows what happens when the rule changes.

Case A GDPR breach notification

GDPR Article 33(1) is a compact stress test because it combines an actor, notification action, recipient, awareness trigger, flexible urgency language, a 72-hour limit, a risk-based exception and a follow-up requirement when notification is delayed [20]. The system should preserve those elements rather than reduce the provision to notify within 72 hours.

FIELD	STRUCTURED INTERPRETATION FOR REVIEW
Actor	Controller
Action	Notify the competent supervisory authority
Trigger	Controller becomes aware of a personal data breach
Deadline	Without undue delay and where feasible no later than 72 hours
Exception	Duty is limited where the breach is unlikely to result in risk to rights and freedoms
Follow-up	Reasons accompany notification when the 72-hour point is exceeded
Evidence	Official EUR-Lex source Article 33 paragraph 1 plus document identifier

Synthetic policy comparison

INTERNAL POLICY	SYSTEM REVIEW SIGNAL	WHY
Report confirmed breaches to Legal within 72 hours of confirmation	Potential mismatch	Trigger may be later than awareness recipient is not supervisory authority and risk exception is absent
Immediately escalate suspected breaches to Privacy Office which assesses notification duties	Insufficient evidence	Strong internal escalation but external notification process is not stated
Privacy Office notifies the competent authority without undue delay and where feasible within 72 hours of awareness unless risk exception applies	Likely aligned	Actor delegation trigger timing recipient and exception appear compatible but implementation evidence is still required

Case B EU AI Act provider obligations

Article 16 of the EU AI Act provides a multi-obligation list for providers of high-risk AI systems, including compliance, identification, quality management, documentation, logs, conformity assessment, declaration and registration duties [19]. The demo can show how one numbered article becomes several atomic obligations that share an actor and scope but have distinct actions, evidence and controls.

Change-impact demonstration

1. Load version one of a synthetic regulation and extract a 90-day account-review duty.
2. Map the obligation to an annual internal policy, creating a potential frequency mismatch.
3. Load version two changing the cadence to 60 days and adding privileged service accounts.
4. Show the semantic diff, affected obligation fields, stale policy mapping, responsible owner and required review date.
5. Export a one-page audit package containing both versions, structured differences and reviewer decision.

13 Risk Security and Governance

A useful compliance system must be designed around failure, sensitivity and accountability from the first prototype.

Risk register

RISK	IMPACT	CONTROL
Missed obligation	False sense of coverage	High-recall detector challenge set severity-weighted review
Unsupported field	Invented actor deadline or exception	Exact span requirement schema gate and null state
Bad source version	Analysis of superseded or draft text	Official identifier effective date hash and authority metadata
Cross-reference failure	Incomplete rule meaning	Reference graph unresolved-state blocker and reviewer queue
Similarity mistaken for compliance	Incorrect gap conclusion	Layered comparison dimension checks and human sign-off
Sensitive policy leakage	Confidentiality breach	Local-first mode encryption RBAC and configurable retention
Model or prompt drift	Non-reproducible results	Version pinning regression suite and approval gate
Automation bias	Reviewer accepts plausible output	Evidence-first UI counterexamples and mandatory rationale
Dataset leakage	Inflated evaluation	Document-level splits duplicate checks and frozen hidden set
Regulatory overclaim	Academic and legal credibility loss	Status labels limitations and no autonomous certification

Governance controls

- Role-based access separates source ingestion extraction review approval and administration.
- Every edit creates a new record version and preserves prior values and rationale.
- Model cards record training data intended use limitations thresholds and known failure modes.
- A source register records authority licensing update frequency and legal-status notes.
- Release gates require provenance tests security review and signed evaluation results.
- Incident procedure covers wrong extraction broken citations unauthorized access and stale regulation data.

Ethical boundary

RegulaTrace assists document analysis and workflow triage. It does not determine legal applicability, interpret disputed law, verify real-world control operation or replace qualified counsel. The user interface and exports must state the evidence reviewed, what was not located and which conclusions remain human judgments.

14 Implementation Roadmap

A fourteen-week plan produces a coherent vertical slice, a measured result and a strong live demonstration.

WEEKS	WORK PACKAGE	DEFINITION OF DONE
1 to 2	Foundation	Repository schema source register baseline reproduced and tests running
3 to 4	Corpus and annotation	Guide pilot double annotation adjudication and document splits
5 to 6	Extraction	Rules linear baseline legal encoder and span extraction compared
7 to 8	Evidence graph	Provenance schema validation obligation registry and review states
9 to 10	Policy comparison	Hybrid retrieval NLI constraint checks and cited outcomes
11	Change radar	Two-version alignment material changes and graph impact queue
12	Reviewer interface	Side-by-side evidence correction approval and export
13	Evaluation and red team	Metrics ablations challenge set human timing and failure catalogue
14	Submission and demo	Frozen release report video reproducibility package and rehearsed demo

Team work allocation

ROLE	OWNERSHIP	CROSS REVIEW
Data and annotation lead	Source acquisition guide corpus splits agreement	Reviewed by NLP lead
NLP lead	Baselines modality and span models error analysis	Reviewed by evaluation lead
Platform lead	Schema graph API storage provenance and security	Reviewed by data lead
Evaluation and UX lead	Metrics reviewer interface study demo and documentation	Reviewed by platform lead

Project gates

1. Gate 1: baseline and dataset reproducible from a clean environment.
2. Gate 2: every extracted record resolves to exact source evidence.
3. Gate 3: model improvement survives document-level evaluation and error review.
4. Gate 4: policy and change signals show dimension-level reasons, not only generated prose.
5. Gate 5: live demo works offline on frozen fixtures and exports a complete audit package.

15 Demonstration and Presentation Strategy

The presentation should make the importance visible through a concrete chain of evidence and impact.

Seven-minute live narrative

TIME	SCREEN	MESSAGE
0 00	Official source and version	Compliance starts with knowing exactly which text and version is being analyzed
0 45	Clause structure and extraction	The system preserves hierarchy and compiles one clause into reviewable fields
1 45	Evidence graph	Every field links to source and every obligation links to owner policy control and evidence
2 45	Policy comparison	A similar policy can still fail on trigger timing recipient or exception
4 00	Regulatory revision	A changed deadline immediately identifies stale mappings and responsible owners
5 15	Reviewer correction	The analyst accepts edits or rejects output with a versioned rationale
6 00	Evaluation dashboard	Results include critical misses calibration provenance and time saved
6 40	Final export	The product creates an auditable package not an unverifiable answer

Presentation assets

- One architecture slide matching Figure 1.
- One before-and-after clause decomposition animation.
- One evidence graph with a revision impact highlighted.
- One evaluation slide led by critical-miss and provenance metrics rather than only accuracy.
- One limitations slide that separates implemented validated and proposed components.

Likely viva questions

QUESTION	STRONG ANSWER
Why not just use an LLM	Because the system requires exact evidence versioning validation and controlled failure; an LLM can assist a stage but cannot be the audit trail
What is the novelty	The contribution is the atomic obligation representation plus evidence and impact graph change radar and layered policy comparison
How do you know it works	Document-split task metrics challenge tests calibration provenance checks and a reviewer time study
Is it legal advice	No it is evidence-linked document analysis and triage with human confirmation
Why keep simple models	Recent regulatory IE results show small datasets can favor character n-gram and rule baselines [6]
What happens when uncertain	The system returns unresolved or insufficient evidence and routes the record for review

16 Limitations Future Work and Conclusion

The most convincing conclusion is ambitious about the system and conservative about the evidence.

Current limitations

The demonstrated baseline remains small, synthetic and sentence-level. It does not establish real-document generalization, extraction accuracy, OCR reliability, cross-reference resolution, policy-comparison quality, latency, security or user benefit. The target design also assumes English and a limited set of source families. Legal ambiguity cannot always be eliminated by more context or a larger model.

High-value extensions

- Multilingual extraction and jurisdiction-specific schema extensions.
- Definition and cross-reference resolution across an entire legal corpus.
- Control-library mappings to frameworks such as NIST CSF with explicit legal-versus-guidance status.
- Event ingestion from regulatory feeds and continuous impact monitoring.
- Active learning that selects high-value uncertain examples for annotation.
- Counterfactual policy drafting that proposes the smallest change needed for review while preserving source citations.
- Formal LegalRuleML export for selected obligation patterns and rule conflict experiments.

Conclusion

RegulaTrace reframes compliance extraction as an evidence-preserving compilation and change-intelligence problem. The project begins with a transparent lexical baseline but advances toward a hybrid system that can represent duties as atomic, validated records; connect them to policies, controls and evidence; compare material constraints; detect the impact of revisions; and stop when evidence is insufficient. Its value is not an automated legal verdict. Its value is a faster, more complete and more defensible path from difficult legal text to accountable human action.

The strongest academic submission will implement one narrow path end to end, measure it honestly and make every output traceable. That combination of ambition, system thinking and evidence discipline is what can make the project stand out.

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A Appendix Project Status and Evidence Plan

This appendix prevents proposal language from being mistaken for completed implementation.

COMPONENT	CURRENT EVIDENCE	NEXT EVIDENCE REQUIRED
Five way lexical classifier	Reported executable baseline and 30 synthetic labels	Repository tests clean run and frozen output
Heuristic field extractor	Reported sample actor action timing condition exception	Span-level evaluation and adversarial fixtures
Document ingestion	Architecture only	Two formats hierarchy preservation and exact source links
Learned modality model	Research design only	Document-split comparison with simple baselines
Evidence graph	Schema and design in this report	Persistent records validation and graph queries
Policy comparator	Algorithm and case study design	Labeled paired set retrieval and outcome metrics
Change radar	Workflow design	Two-version demo and impact recall
Reviewer interface	UX requirements	Working correction approval and export flow
Human benefit	Hypothesis	Timed comparative study and workload feedback

Definition of a defensible final claim

At submission, every statement about implemented capability should point to a test, screenshot, dataset version, code commit or measured result. Planned modules remain labeled planned. The project becomes impressive because it demonstrates disciplined engineering and research, not because it hides limitations.

B Appendix Adversarial Challenge Set

The challenge set targets linguistic patterns most likely to create confident but unsafe errors.

CHALLENGE	EXAMPLE PATTERN	EXPECTED BEHAVIOUR
Separated negation	No record shall be disclosed	Prohibition not obligation
Epistemic may not	Records may not be complete	Information not prohibition
Quantifier scope	Not every contractor must train	Do not infer universal duty
Implicit necessity	Access rights need to be reviewed	Obligation candidate
Passive actor	Notice must be provided by the controller	Actor extracted from by phrase
Inherited actor	Heading Duties of Providers followed by shall keep logs	Actor inferred with heading support
Nested exception	Must notify unless low risk except where law requires	Preserve both exception scopes
Event anchored deadline	Within 72 hours after awareness	Capture duration and awareness trigger
Recurring cadence	At least once in every rolling 90 day period	Normalize recurrence and lower bound
Definition dependency	Significant incident as defined in Article 2	Store and resolve definition link
Cross reference	Subject to paragraph 4	Retrieve paragraph 4 before confirmation
Table obligation	Required values expressed across row and header	Preserve table context
Quoted modality	Policy states users shall change passwords	Avoid treating descriptive quotation as issuer duty
Conflicting provisions	General rule and later specific exception	Route conflict for review
Version change	90 days becomes 60 days	Material tightening and affected policy queue

End of report